

Raghav Senthil Kumar

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EDUCATION

University of Illinois Urbana-Champaign

Champaign, IL

B.S. in Computer Science and Statistics

Expected: May 2029

- **Coursework:** Computer Science I (Java), Computer Science II (C++), Data Science & ML (Python), Calculus III
- **Activities:** Disruption Lab, SIGRobotics, AI Alignment, Agentic AI, BuildIllinois, Innovation LLC

EXPERIENCE

Founding Software Engineer | Clusion

September 2025 – January 2026

- Built the document ingestion and search pipeline (*Python FastAPI*) for a competitive high school debate research SaaS platform on *AWS*, collaborating with an *Agile* team of **10 engineers** and serving **10K+ site visitors**
- Indexed **600K+ debate-evidence documents** on *Elasticsearch* via *FastAPI*, returning results in **under 1 second**
- Applied data validation in a text-retrieval algorithm that checks source URLs and filters statistical outliers with **IQR**
- Flagged **30%** of evidence documents as identical duplicates with a de-duplication algorithm, reducing index bloat

Product Development Intern | Jetson

January 2025 – February 2025

- Conducted **7 iterations of UI/UX and user acceptance testing** on 3 features of the Jetson App (100K+ users)
- Delivered a **5-page report** recommending strategies to optimize user engagement directly to Jetson's CEO
- Redesigned onboarding on *Figma*, cutting it from **12 to 5 pages**, validated via **A/B testing** on 30 test users

Data Analytics Intern | STEM E

June 2024 – July 2024

- Built reports from 3 years of TikTok data (**200+ posts**) with *Python (pandas)*, *Power BI* and *Microsoft Excel*
- Identified that “science fun facts” posts comprised 69.3% of views, despite accounting for 7% of all content
- Pitched a “fun facts” strategy to the Executive Director, who adopted it; engagement rose **24% (8K+ views)**

PROJECTS

Citation Chain | [Live Website](#) | [Source Code](#)

June 2026

- Developed an *ETL* data pipeline that ingests papers via *OpenAlex* & backfills metadata from *Semantic Scholar*
- Stored paper data on *Supabase (PostgreSQL)* and wrote *SQL* queries to filter papers by field, year and citation count
- Built an interactive citation graph (*React Flow*) that maps a paper's outbound citations across **26 research fields**
- Deployed a **RAG agent (Claude Sonnet 4.5)** that retrieves session-scoped vector documents (*Supernemory*)

SchoolPulse AI | [Live Website](#) | [Source Code](#)

June 2026

- Engineered a **suite of 3 edge AI tools** that collects data regarding a school's energy, water and food consumption
- Built a *Python FastAPI* backend that ingests live water, waste & energy data from all 3 tools, runs **4 Random Forest models** to detect anomalies & deliver actionable steps via **Gemma Voice LLM** on a *Next.js* dashboard
- Led a team of 5 across project ideation, development and presentation. **Finalist at USAII Global Hackathon**

Truck Pricing VLM | [Live Website](#) | [Source Code](#)

September 2026

- Led a team of 4 on a pricing algorithm for **Kamion (YC S22)**. **2nd Place at UIUC Founder's Hackathon**
- Scraped **15,000** truck-listing images, auditing data quality via de-duplication and non-USD currency filtering
- Designed a *JSON schema* for a **Claude Sonnet 4.6 agent** to extract condition, brand, model and year from listing photos, revising it after each wrong prediction until **10 consecutive images** were correct on each of **3 features**

EcoBin | [Research Paper](#) | [Source Code](#)

March 2026 – May 2026

- Designed a two-stage *EfficientNetV2-S* neural network (**4.39M** parameters) in *PyTorch*, trained on **15K+ images** to classify waste into 4 disposal pathways at **96.13% accuracy**. Published a paper on *arXiv* and 2 datasets on *Kaggle*
- Built a synthetic data generation pipeline using *U2-Net* segmentation and alpha-compositing to generate 9K images to train the model to detect contamination in recyclable waste. **Placed 2nd at the Arizona State Science Fair**
- Engineered a smart bin prototype using *Arduino Uno* that sorts the waste item with **less than 2 second latency**

SKILLS

Programming Languages: Python, SQL, JavaScript, HTML/CSS, Java, C++

Frameworks/Platforms: AWS, PostgreSQL, MySQL, Elasticsearch, Supabase, FastAPI, Flask, Firebase, Google Cloud

DevTools: Git, GitHub, Docker, CI/CD, VS Code, Claude Code, Kaggle, Google Collab, Jupyter Notebooks, Catch2

Data & Analytics: Power BI, Tableau, Microsoft Excel, pandas, NumPy, matplotlib, A/B Testing, Hootsuite

Concepts/Practices: Data Analysis, ETL/ELT, Statistical Modelling, Data Pipelines, Testing, Quality Control